A Stepwise Refining Image-Level Weakly Supervised Semantic Segmentation Method for Detecting Exposed Surface for Buildings (ESB) From Very High-Resolution Remote Sensing Images

Xin Huang, *Senior Member, IEEE*, Wenrui Wang, Jiayi Li, *Senior Member, IEEE*, Leiguang Wang, and Xing Xie

***Abstract*— Exposed surface for buildings (ESB), which refers to exposed surfaces with traces of building construction, often leads to urban dust. Accurate ESB detection is important for planning urban development and improving urban environment. Fine-grained monitoring of ESB typically needs massive high-quality pixel-level labels, which are demanding and expensive. In contrast, obtaining cost-efficient image-level labels is more promising. Most image-level weakly supervised methods can extract pixel-level pseudo labels using the class activation map (CAM) generated by the classification network. Subsequently, these labels are applied to train the semantic segmentation network. However, the CAM is easy to miss fine-grained information, which leads to label noise. Moreover, the downsampling in the segmentation networks will further loss the spatial information. Furthermore, the sparse distribution and irregular shape of ESB pose additional challenges. Given these problems, we propose a stepwise refining image-level weakly supervised semantic segmentation method (SRIWS): 1) we introduce a new data augmentation method called SRMix to oversample the classification dataset; 2) we propose a two-branch network with a superpixel pooling layer (SPNet) as the semantic segmentation network to capture both global semantic information and spatial details; and 3) to alleviate the impact of potential noise in the initial labels, we design the high-confidence sample filtering operation (HSF) during the SPNet training. The evaluation experiments for the SRIWS were performed on three datasets. The results confirm that our proposed SRIWS presents a superior performance in recognizing ESB compared with existing state-of-the-art methods. In addition, numerous ablation experimental results indicate the effectiveness and robustness of our SRIWS.**

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Xin Huang, Wenrui Wang, and Jiayi Li are with the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan 430079, China (e-mail: xhuang@whu.edu.cn; 2017302590121@whu.edu.cn; zjjercia@whu.edu.cn).

Leiguang Wang is with the Institute of Big Data and Artificial Intelligence and the Key Laboratory of State Forestry and Grassland Administration on Forestry and Ecological Big Data, Southwest Forestry University, Kunming 650024, China.

Xing Xie is with the School of Environmental and Mapping Engineering, Suzhou University, Suzhou 234000, China.

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***Index Terms*— Exposed surface for buildings (ESB), image-level samples, very high-resolution remote sensing (VHR) images, weakly supervised semantic segmentation.**

I. Introduction

SINCE the 21st century, the process of urbanization has been accelerating at an unexpected speed [1], [2], driving the rapid growth of built-up areas [3], and leading to significant land use changes [4]. As the built-up areas generally start from the exposed surface for buildings (ESB), it is meaningful to consider the distribution of ESB in the investigation of the land use change in cities. In addition, the rapid development of urbanization comes with extensive urban construction [5], and the numerous ESB generated during urban construction may lead to serious urban dust [6], [7]. Thus, accurate ESB identification is crucial for guiding urban planning and improving urban environment.

Since the 1990s, the rapid development of high-resolution sensor technology has provided data support for fine-grained monitoring of ESB [8]. One of the key techniques for very high-resolution remote sensing (VHR) image interpretation is semantic segmentation [9], which is defined as assigning every pixel in the image with a land-cover label, such as ESB or Non-ESB [10]. Ideally, high-precision semantic segmentation operations require massive high-quality pixel-level annotations (also known as "strong" labels) [11], which is very demanding and laborious [12]. In contrast, the cost-efficient weak labels [13], [14] are relatively practical, although they contain less information [13]. There are several types of weak labels, e.g., point labels [15], bounding boxes [16], [17], scribbles [18], [19], and image-level labels [20], [21], [22], [23]. As suggested from recent reviews of semantic segmentation on the basis of weak labels [11], [24], image-level labels are more promising. Therefore, this study focuses on utilizing image-level weakly supervised segmentation methods to detect ESB from the VHR images.

Most image-level weakly supervised segmentation methods can be partitioned into two main steps. First, by thresholding class activation map (CAM) estimated from an image clas-

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sification network, a set of initial pixel-level pseudo labels can be obtained. Then, a segmentation network is optimized to classify the land-cover class for each test pixel using these initial labels as supervision [22]. When applying current weakly supervised studies to VHR images, there are two main limitations.

First, CAM estimated from the image classification network aims to reflect the major semantic of the sample, missing the fine-grained (i.e., pixel level) spatial details, and resulting in label noise [25]. This problem will be further worsened when the size of ground objects displays significant differences [20]. To tackle this problem, some existing studies have paid their attention to optimize the detail representation capability of CAM during image classification network training [20], [26]. Nevertheless, the aforementioned methods may not be suitable when dealing with the segmentation task in an unbalanced foreground-to-background ratio. Other studies utilized deep learning techniques with label noise to alleviate the poor influence of the noisy labels, including noise modeling technique [27], noise-robust loss functions [28], [29], and noisy label correction [30], [31], [32], [33], [34]. While deep learning techniques with label noise have shown positive effects to some extent, eliminating the negative effect of noisy labels remains challenging.

Second, the initial pixel-level pseudo labels lack the boundary information of ground object, and the process of downsampling in the semantic segmentation networks, such as UperNet [35], FarSeg [36], PointFlow [37], and SegNeXt [38], will further cause the lack of spatial information [31]. To address this problem, most existing studies used the superpixel map [26], [31] or conditional random field (CRF) operation [22], [39] to obtain refined object boundaries. However, CRF often suffers from the difficult selection of spatial smoothing scales when combining unary and binary potentials [40], [41], while superpixel map often tends to oversegment in order to guarantee the purity of superpixels, making it difficult to reflect the global information of the current image within a single superpixel [42], [43].

Attention should be drawn to the fact that considering the sparse spatial distribution [see Fig. 1(a) and (b)] and irregular shape [see Fig. 1(c)–(e)] of ESB, its semantic segmentation task puts higher requirements on weakly supervised learning with image-level labels. To tackle the aforementioned issues, we designed a stepwise refining image-level weakly supervised semantic segmentation method (SRIWS) to identify the ESB from the VHR images. The SRIWS mainly involves two steps. To begin, we utilized the image-level labels as supervision to train an image classification network, resulting in the initial pixel-level pseudo labels. In addition, we proposed a new sample augmentation strategy, termed semantic relatedness mixing (SRMix), during the image classification network training. The SRMix aims to enlarge the ESB samples and mitigate the impact of the class imbalance. Subsequently, we proposed a two-branch network with a superpixel pooling layer (SPNet), which can simultaneously maintain the global semantic information and the spatial details provided by superpixels. Considering the problem that the initial labels may be noisy, we designed the high-confidence sample filtering operation (HSF) to select samples for iterative correction of initial labels during the SPNet training. Once the label was corrected well, the fully corrected pixel-level pseudo labels were utilized to fine-tune the SPNet and output the final ESB extraction results.

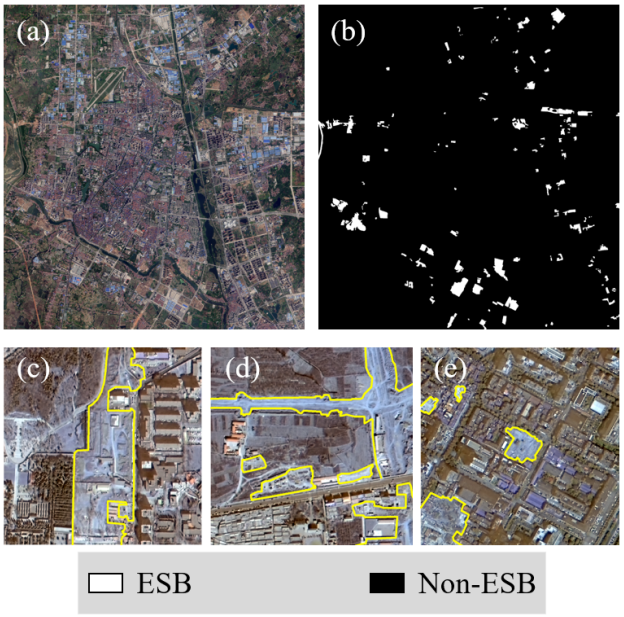

Fig. 1. Visualization of ESB. (a) VHR image in Xiaogan. (b) Pixel-level label corresponding to (a). In(a) and (b), it is evident that compared with the diverse and complex background, ESBs solely occupy comparatively small areas, which potentially leads to the class imbalance problem. (c)–(e) VHR images of ESB, with the yellow line representing the boundary between ESB and the surrounding ground objects. From (c) to (e), it can be seen that ESBs have different sizes and various shapes. Therefore, it is important to design an effective segmentation network that is capable of preserving sufficient spatial details.

The novel aspects of our study are summarized as follows.

1) The SRMix method is proposed during the image classification to weaken the negative impact of the class imbalance. Compared with other data augmentation methods, the SRMix synthesizes pseudo samples by extracting key information from the original samples, suppressing label noise during the data augmentation.
2) A two-branch SPNet is proposed to alleviate the blurry boundary problem. Without adding additional computational cost, the SPNet takes full advantage of the spatial details provided by the superpixels and the semantic information extracted by the network itself. Moreover, an HSF is designed for reducing the influence of noisy labels during the SPNet training.

The remaining sections of our study are structured as described in the following. Section II presents the created datasets utilized for our study. Section III describes our SRIWS in detail. Section IV displays the experiments and the ablation study is introduced in Section V. In conclusion, Section VI provides a summary.

II. Datasets

To investigate the effectiveness of the SRIWS, three datasets, namely, Google Earth ESB detection dataset

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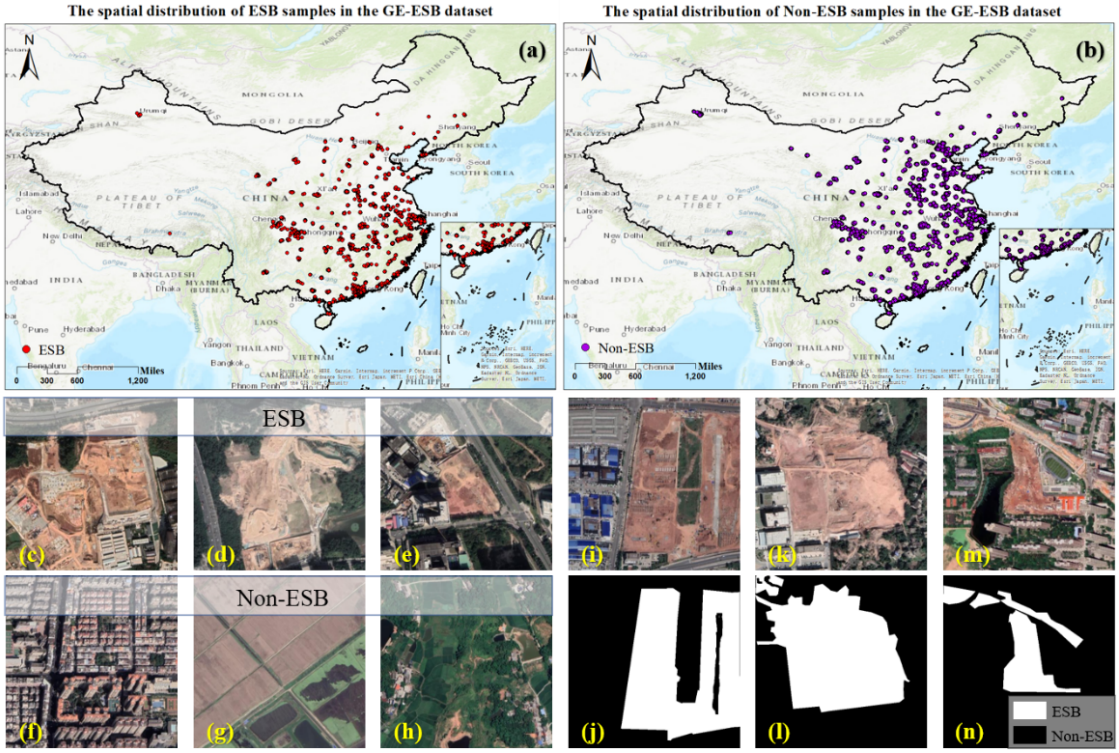

Fig. 2. Illustration of GE-ESB. (a) Sample distribution of ESB class in the GE-ESB dataset. (b) Sample distribution of Non-ESB class in the GE-ESB dataset. (c)–(e) ESB images within the image classification samples. (f)–(h) Non-ESB images within the image classification samples. (i)–(n) ESB images and the corresponding pixel-level labels within the image segmentation samples.

(GE-ESB), Gaofen-2 ESB detection dataset (GF2-ESB), and a large-area (city-level) test dataset, were created.

A. Google Earth Exposed Surface for Buildings Detection Dataset

The GE-ESB dataset was collected from Google Earth data downloaded from https://www.google.com in September 2021. Images in the GE-ESB dataset, whose spatial resolution is 1 m, consist of RGB visible bands. Fig. 2 shows the sample spatial distribution and sample visualization information of the GE-ESB dataset.

The GE-ESB dataset can be partitioned into two parts: an image classification dataset and an image segmentation dataset. Each sample in the GE-ESB dataset contains an image with a size of 512 × 512 × 3 and a group of corresponding labels. The image within the image classification sample corresponds to a single class, namely, ESB or Non-ESB. In this article, an image is identified as an ESB sample if it contains ESB objects larger than 100 × 100 m. The image size in the datasets used in this article is 512 × 512 pixels with a spatial resolution of 1 m. Therefore, if the proportion of ESB pixels in an image exceeds 3.8%, it is considered an ESB sample. Images that do not contain ESB pixels are categorized as Non-ESB samples. We chose China as the study area to collect ESB samples of the image classification dataset, while Non-ESB samples were selected based on the guideline of spatial uncorrelatedness. The spatial uncorrelatedness of samples refers to defining the minimum spatial distance as the width of the samples. In this way, a total of 3000 ESB samples and 27 661 Non-ESB samples were collected. To obtain the training, validation, and testing image sets for the image classification dataset, we divided each class using a ratio of 6:2:2. The image segmentation dataset, which is only used for testing, consists of 80 ESB samples, with each pixel in images corresponding to an individual label, that is, ESB or Non-ESB. The above annotation process was done manually by three experienced remote sensing image interpreters.

B. Gaofen-2 Exposed Surface for Buildings Detection Dataset

Meanwhile, we utilized Gaofen-2 satellite images as a data source to create the GF2-ESB dataset. The GF2-ESB dataset was acquired from the China Centre for Resources Satellite Data and Application (https://www.cresda. com). We selected 12 relatively economically developed cities in China as the study area due to the large number of ESBs available in these regions. From the GF-2 images of the study area, we chose images with less than 10% cloud coverage. Through manual selection, we finally selected 28 sets of GF-2 images. The collection time is shown in Table I, and the distribution of the 28 image sets and the sample visualization information are shown in Fig. 3. Each image set contains a panchromatic image and a multispectral image. We carried out a series of sequential procedures on all the sets during the preprocessing process. Specifically, we applied radiometric correction and atmospheric correction to the multispectral images, as well as radiometric correction to the panchromatic images. Then, we implemented orthorectification and image-to-image

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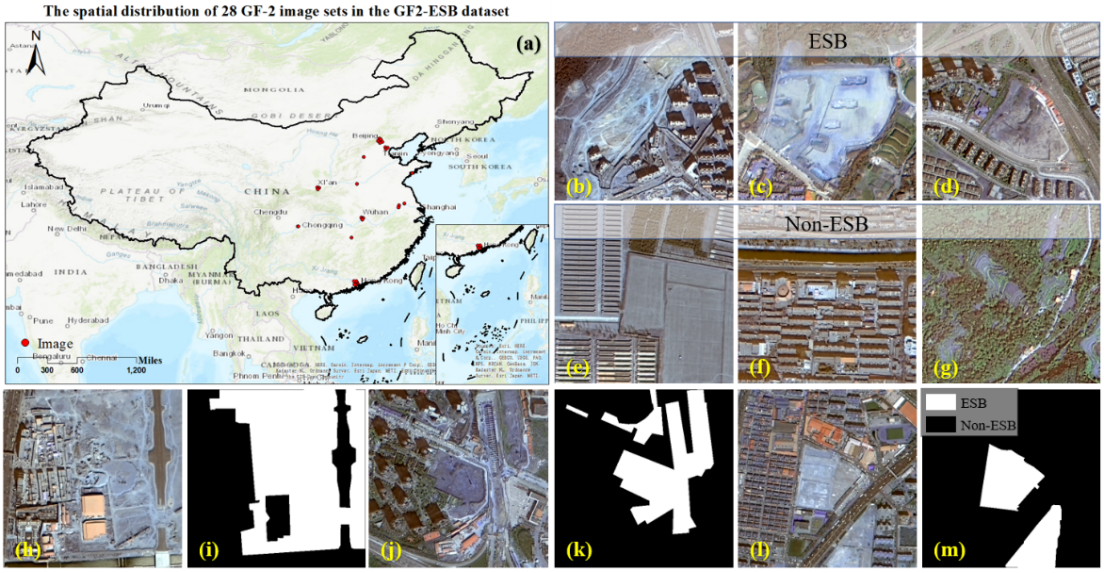

Fig. 3. Illustration of GF2-ESB. (a) Image distribution of 28 GF-2 image sets (the distribution of ESB samples and Non-ESB samples is the same). (b)–(d) ESB images within the image classification samples. (e)–(g) Non-ESB images within the image classification samples. (h)–(m) ESB images and the corresponding pixel-level labels within the image segmentation samples.

TABLE I
COLLECTION TIME OF THE GF2-ESB DATASET

City	Time	# images
Zhengzhou	2018-04-16	1
Shijiazhuang	2015-03-04	1
Tianjin	2015-06-20	3
Changsha	2018-03-22	1
Nanjing	2017-10-09	2
Qingdao	2018-04-19	1
Chongqing	2018-04-02	1
Beijing	2015-02-17	4
	2015-02-12	1
Wuhan	2016-09-01	2
	2016-12-08	1
Xian	2016-11-30	1
	2017-03-08	2
Yangzhou	2018-04-19	1
Shenzhen	2017-10-29	4
	2017-02-15	2

registration on all the images. Finally, we generated multispectral images with a higher spatial resolution (i.e., 1 m) by fusing the panchromatic images at 1 m with the multispectral images at 4 m. Visible (red, blue, and green) and near-infrared bands were used in our study.

Similar to the GE-ESB dataset, the GF2-ESB dataset can be partitioned into an image classification dataset and an image segmentation dataset. We cropped from the 28 sets of images to create the image classification dataset and obtained a total of 3000 ESB samples and 9200 Non-ESB samples. The generation of image-level labels is the same as the GE-ESB dataset. Each sample in the GF2-ESB dataset is sized at 512 × 512 pixels. To obtain the training, validation, and testing image sets for the image classification dataset, we divided each class using a ratio of 6:2:2. The image segmentation dataset, which is only used for testing, consists of 80 ESB samples.

C. Large-Area (City-Level) Test Dataset

The large-area (city-level) test dataset was obtained from Google Earth data in October 2021. Images in the large-area (city-level) test dataset, whose spatial resolution is 1 m, contain RGB visible bands. The dataset consists of two city-level samples, with each pixel in images corresponding to an individual label. One image is in Xiaogan sized at 10 696 × 11 520 pixels and the other is in Foshan sized at 10 696 × 10 741 pixels. Fig. 4 shows the visualization of the dataset. Compared with the image segmentation samples in the GE-ESB and GF2-ESB datasets, the large-area (city-level) test dataset has larger coverage and covers more diverse ground objects. Therefore, it can be used to test the effectiveness of the SRIWS in scenarios with sparse and unbalanced ESB samples.

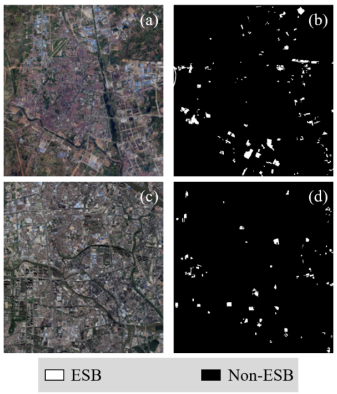

Fig. 4. Large-area (city-level) test dataset. (a) and (b) VHR image in Xiaogan and its corresponding pixel-level labels, respectively. (c) and (d) VHR image in Foshan and its corresponding pixel-level labels, respectively.

III. METHOD

Our proposed SRIWS contains the subsequent two steps.

1) Initial pixel-level pseudo labels generation based on the SRMix: our proposed SRMix is an oversampling technique to strengthen the ESB samples in the image classification dataset. The classification network was

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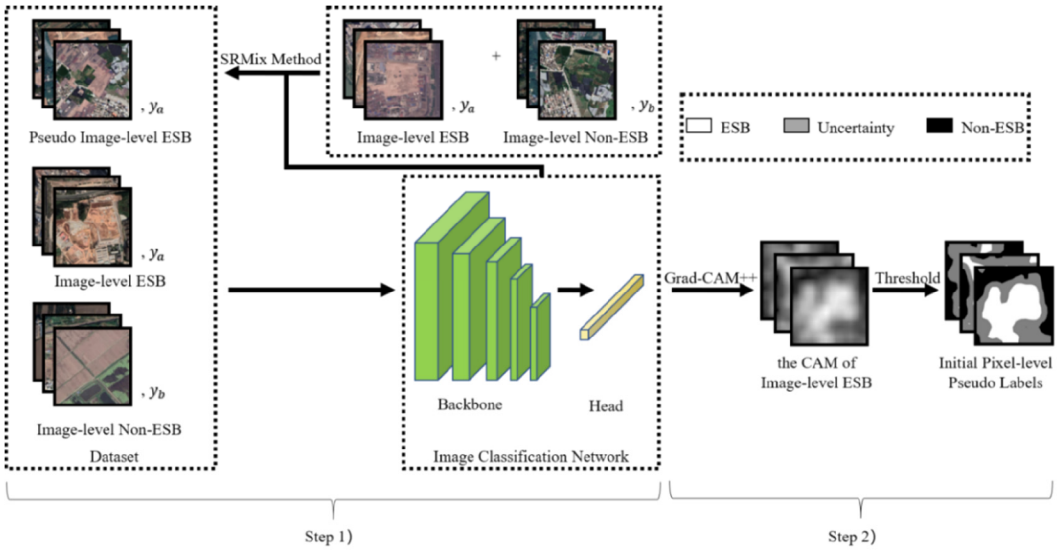

Fig. 5. Workflow of initial pixel-level pseudo labels generation based on the SRMix. Step 1) trained the image classification network using the original image classification dataset for warm-up and subsequently designed the SRMix to generate the pseudo classification dataset as another input to the network. After the network training, step 2) used Grad-CAM++ to generate the CAM of the ESB images and then used the threshold segmentation method to extract the initial pixel-level pseudo labels from CAM. The label corresponding to the image-level ESB sample is y_a, and the label corresponding to the image-level Non-ESB sample is y_b.

trained using both the original samples and pseudo samples generated by the SRMix. Subsequently, the CAM and the initial pixel-level pseudo labels were generated from the classification network.

2) ESB semantic segmentation based on the SPNet: our proposed SPNet adopted iterative correction for the initial pixel-level pseudo labels. During the pseudo label correction, we designed the high-confidence sample filtering operation to remove the potential noise in the initial labels. Once the label was corrected well, the SPNet was fine-tuned utilizing the fully corrected pixel-level pseudo labels as supervision and output the final ESB extraction results.

A. Initial Pixel-Level Pseudo Labels Generation Based on the SRMix

In our study, we used image-level samples (i.e., ESB samples and Non-ESB samples) to train the classification network and generated pixel-level pseudo labels that can provide the initial spatial position information of ESB. The workflow is presented in Fig. 5. While optimizing the image classification network, the SRMix was proposed to mitigate the class imbalance resulting from the uneven number of ESB samples and Non-ESB samples. The initial pixel-level pseudo labels generation based on the SRMix contains two steps: 1) image classification network training based on the SRMix and 2) initial pixel-level pseudo labels generation based on CAM.

1) Image Classification Network Training Based on the SRMix: Due to the sparse spatial distribution of ESB, the number of image-level ESB samples is generally much less than Non-ESB samples. When utilizing the imbalanced classification dataset, the image classification network may exhibit a bias to the Non-ESB class and disregard the ESB class [44], [45], resulting in the performance degradation of the network [46], as it may struggle to accurately classify samples from the minority class. To solve this problem, the SRMix mixed ESB and Non-ESB samples using semantic information to generate new pseudo-ESB samples, thereby reducing the disparity in the sample numbers between classes as well as alleviating the impact of class imbalance. The workflow of the SRMix is shown in Fig. 6.

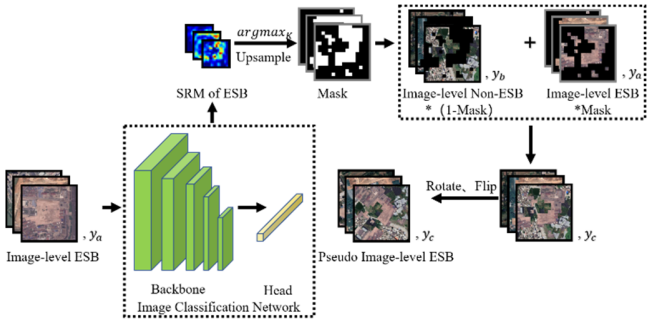

Fig. 6. Workflow of the SRMix. The image-level ESB samples were input to our image classification network, which would output the SRMs of samples to highlight their important regions. The K pixels with the largest values in the SRMs were given the value of 1, while the other pixels were given the value of 0. Via upsampling operation, binary masks consisting of 0 and 1 with the same size as the ESB samples were obtained. The masks describing where to apply the dropout and fill-in were used to mix the ESB samples and Non-ESB samples, and the mixing results were then rotated and flipped to obtain the new pseudo ESB samples. The label corresponding to the image-level ESB sample is written as y_a, the label corresponding to the image-level Non-ESB sample is y_b, and the label corresponding to the generated pseudo ESB sample is y_c.

The principle of CAM was initially presented in [47]. CAM is a class activation heatmap with a certain localization ability generated from an image classification network [48], which can be used to interpret how a region correlates with an image-level label [49]. In this study, we estimated the semantic composition of ESB samples based on their CAMs, which served as the guiding information for sample mixing.

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Given an image I, calculate its class activation map $\mathrm{CAM}_c(x, y)$ for the class c at spatial position (x, y). $\mathrm{CAM}_c(x, y)$ represents the relevance between spatial position (x, y) and class c, with a larger value indicating a stronger association. To further normalize $\mathrm{CAM}_c(x, y)$ to sum to one, a quantized semantic relatedness map (SRM) was obtained. The size of the SRM remains equal to the feature map derived from the last convolutional layer in the image classification network, that is, a pixel of the SRM corresponds to a rectangular neighborhood with a size of $m \times n$ in the original image I

$$\begin{cases} m = \mathrm{Height}(I)/\mathrm{Height}(\mathrm{SRM}(I)) \\ n = \mathrm{Width}(I)/\mathrm{Width}(\mathrm{SRM}(I)) \end{cases} \tag{1}$$

where Height and Width, respectively, represent the length and width of the images.

The SRMix is described in the following manner: given an ESB sample, denoted as I_a, we calculated its SRM as $\mathrm{SRM}(I_a)$, cropped the area in I_a corresponding to the K pixels with the largest values in $\mathrm{SRM}(I_a)$, and then pasted it to the corresponding place in any Non-ESB sample I_b. After rotating, flipping, and other operations, a new pseudo sample I_c could be obtained. The new sample I_c would be assigned as the label of I_a. The above method can be formulated as

$$\begin{cases} I_c = \mathrm{Augument}(\mathrm{Mask} * I_a + (1 - \mathrm{Mask}) * I_b) \\ y_{I_c} = y_{I_a} \end{cases} \tag{2}$$

where y represents the image-level label of the sample, i.e., ESB or Non-ESB, and Mask is a mask filled with 0 and 1 with the same size as the ESB sample. The one-valued region in Mask is the region corresponding to the K pixels with the largest values in $\mathrm{SRM}(I_a)$, and the remaining are all zero-valued regions. Augument represents rotation, flipping, and so on. The value of K satisfies the formula

$$K = \mathrm{int}(\mathrm{att_grid} * \mathrm{lam})\frac{1}{4} \leq \mathrm{lam} \leq \frac{3}{4} \tag{3}$$

where att_grid is the product of the length and width of the $\mathrm{SRM}(I_a)$, $\mathrm{lam} \sim \mathrm{Beta}(\alpha, \alpha)$.

In order to generate more reasonable SRM, the image classification network was first trained using the original classification dataset for warm-up. The pseudo classification dataset was then generated using the SRMix and input to the classification network along with the original dataset for subsequent training.

2) Initial Pixel-Level Pseudo Labels Generation Based on CAM: Once the image classification network was trained well, we utilized the Grad-CAM++ [50] to produce CAMs with localization information and then generated initial pixel-level pseudo labels based on the CAMs. The process of generating initial labels is as follows. Given the thresholds t_1 and t_2, if the CAM value of a pixel exceeded t_1, the pixel was assigned as the foreground (i.e., ESB); if the CAM value of a pixel was less than t_2, the pixel was assigned as the background (i.e., Non-ESB); and the rest of pixels were assigned as uncertain class.

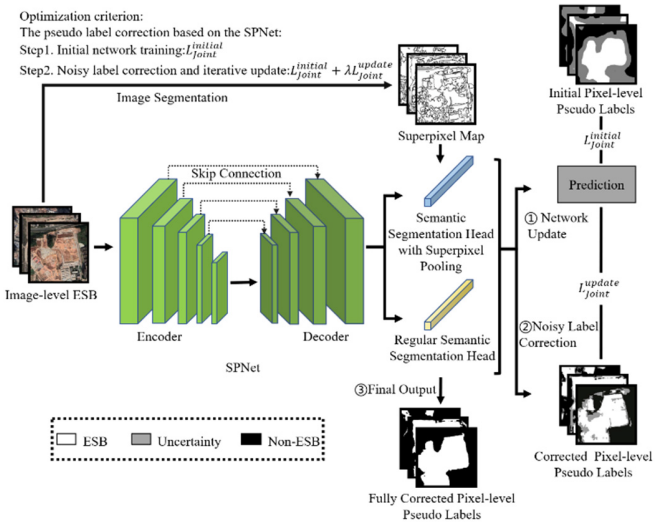

Fig. 7. Workflow of pseudo label correction based on the SPNet. We used the image segmentation technique to generate superpixel maps of image-level ESB samples and fed them into the SPNet together with the ESB samples. During the initial network training stage, we used the initial pixel-level pseudo labels as the supervision to optimize the SPNet. During the NLC and iterative update stage, we used the initial labels as well as the corrected pixel-level pseudo labels together to optimize the SPNet and iteratively performed NLC and network update until the full corrected pixel-level pseudo labels were obtained. The legend of the initial labels and the corrected labels are presented in the bottom-left corner of 7, and the full corrected labels contain only the ESB pixels and the Non-ESB pixels.

B. ESB Semantic Segmentation Based on the SPNet

When weakly supervised learning with image-level labels is applied to the processing of remote sensing image data, there is a problem that the boundaries of ground object are difficult to retain effectively. Specifically, for the task of ESB semantic segmentation, the irregular shape of ESB increases the difficulty of its boundary extraction. Superpixels are small regions consisting of groups of pixels at adjacent locations with similar features, e.g., color, brightness, and texture [51], which can effectively constrain the object boundary [52] and provide prior knowledge of object shape [26]. However, the oversegmentation usually leads to difficulty in reflecting the global information of the current image in a single superpixel. With the consideration mentioned above, a two-branch SPNet was designed to use both global semantic information and spatial detail information. Furthermore, considering the potential noise in the initial pseudo labels, we designed the HSF during the SPNet training. The ESB semantic segmentation based on the SPNet contains two steps: 1) pseudo label correction based on the SPNet and 2) ESB semantic segmentation.

1) Pseudo Label Correction Based on the SPNet: The workflow of pseudo label correction based on the SPNet is shown in Fig. 7.

a) Network structure: The SPNet consists of two components: 1) a feature encoder–decoder structure $f_{\mathrm{enc-dec}}$ and 2) two semantic segmentation heads f_{seg} and f_{sup}. As shown in Fig. 7, the SPNet used the Unet encoder–decoder structure as its main body to derive the feature maps that have the equal size as ESB samples. Subsequently, two branches were utilized to predict the pixel-level results. The first branch used the

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regular semantic segmentation head f_{seg} to output the semantic segmentation predictions pixel by pixel; the second branch used the semantic segmentation head with the superpixel pooling layer f_{sup} to output the semantic segmentation predictions object by object. To generate the necessary superpixel maps for the SPNet, we used a graph-based image segmentation algorithm [53] and subsequently aggregated superpixels with similar features to produce the final superpixels.

The principle of the superpixel pooling layer shares the similarity with the regular pooling layer, that is to say, the local regions are aggregated using a simplifying function [e.g., max(·) or avg(·)] [54]. For the regular pooling layer, the aggregated local regions are predefined square neighborhoods; for the superpixel pooling layer, the aggregated local regions are determined by the superpixel map of the input image, thus reducing the image information to each feature vector corresponding to a superpixel. In the SPNet, given an image I, the feature map $F \in \mathbb{R}^{H\times W\times C}$ was obtained through the encoder–decoder structure $f_{\text{enc-dec}}$. Then, F was input to the subsequent superpixel pooling layer, the output $\text{Sup}_{N,K}$ was obtained as shown in (4), and $\text{Sup}_{N,K} \in \mathbb{R}^{N\times C}$

$$\text{Sup}_{N,C} = \max F_{i,C} | S_i = n \tag{4}$$

in which H refers to the length of feature map F, W represents the width, C is the channel number of F, N is the total number of superpixels in the superpixel map of image I, and S_i represents the set of all pixels in the nth superpixel.

The receptive fields of superpixels and their regular square neighbors are located close with differences at the edges. As a result, the outputs of the two branches tend to be similar but hardly reach complete agreement. The two-branch structure is used to provide additional spatial information during the SPNet training, which helps to achieve excellent performance improvement of the network. At the same time, the SPNet shares the same encoder and decoder, avoiding the excessive computational cost caused by the two-branch structure.

b) Training steps: It consists of two stages—Step 1: initial network training and Step 2: noisy label correction (NLC) and iterative update.

Step 1 (Initial Network Training): Existing studies provide evidence that neural networks prefer to learn from pure samples during the early training and gradually introduce more diverse samples as the training progresses [55]. Consequently, the networks are prone to learning mislabeled samples in the later stages. In addition, deep networks are less likely to fit mislabeled samples with a high learning rate [56]. Based on the above studies, using the initial pixel-level pseudo labels as supervised information, the SPNet was optimized for several epochs under a high learning rate.

Step 2 (NLC and Iterative Update): For the outputs of two semantic segmentation heads in the SPNet, we assigned the corresponding labels to the pixels with the same prediction results and the uncertain label to the pixels with different prediction results to obtain the corrected pixel-level pseudo label. The uncertain regions were not used in the subsequent network update training. At this time, given that the majority of initial pseudo labels are correct, the SPNet used the initial labels and the corrected labels as supervised information for network update training.

c) Fully corrected pixel-level pseudo labels: When the network update was completed, we used the prediction results generated by the regular semantic segmentation head as the fully corrected pixel-level pseudo label (shown in Fig. 7) for the subsequent ESB semantic segmentation task. At this point, the fully corrected pixel-level pseudo labels no longer contain any uncertain class.

d) Optimization criterion: The optimization criterion L_{Joint} of the SPNet can be formulated as

$$L_{\text{Joint}}(x_i, y_i) = \left(1-\lambda_1\right) L_{\text{cross}}(x_i, y_i) + \lambda_1 L_{\text{contra}}(x_i) \tag{5}$$

where x_i is the ESB sample and y_i is the pixel-level pseudo label corresponding to x_i. λ_1 is the weight representing the importance of L_{contra}. L_{cross} represents the cross-entropy loss for each of the two branches of the SPNet, which can be expressed as

$$L_{\text{cross}}(x_i, y_i) = -\sum_{i=1}^{n} y_i \log\left(F_{\text{seg}}(x_i)\right) - \sum_{i=1}^{n} y_i \log\left(F_{\text{sup}}(x_i)\right) \tag{6}$$

where n represents the total number of samples used in the SPNet; $F_{\text{seg}}(x_i)$ denotes that the samples x_i were first fed into the feature encoder–decoder structure, followed by regular semantic segmentation of the head; and $F_{\text{sup}}(x_i)$ denotes that x_i were first fed into the feature encoder–decoder structure, followed by the semantic segmentation head with the superpixel pooling layer.

To maximize the consistency between the two branches, we used the co-regularization method, i.e., adding the contrast term L_{contra} as one of the optimization criteria. L_{contra} can be summarized as

$$L_{\text{contra}}(x_i) = D_{\text{KL}}\left(F_{\text{seg}}(x_i) \middle|\middle| F_{\text{sup}}(x_i)\right) + D_{\text{KL}}\left(F_{\text{sup}}(x_i) \middle|\middle| F_{\text{seg}}(x_i)\right)$$

where

$$D_{\text{KL}}(F_{s1}(x_i) || F_{s2}(x_i)) = \sum_{i=1}^{n} F_{s1}(x_i) \log \frac{F_{s1}(x_i)}{F_{s2}(x_i)} \tag{7}$$

where $D_{\text{KL}}(\cdot)$ represents the Kullback–Leibler (KL)-divergence and $F_{s1}(\cdot)$ and $F_{s2}(\cdot)$ are used to denote $F_{\text{sup}}(\cdot)$ or $F_{\text{seg}}(\cdot)$. The relationship between $F_{s1}(\cdot)$ and $F_{s2}(\cdot)$ can be expressed as

$$\begin{cases} F_{s2}(\cdot) = F_{\text{seg}}(\cdot), & \text{if } F_{s1}(\cdot) = F_{\text{sup}}(\cdot) \\ F_{s2}(\cdot) = F_{\text{sup}}(\cdot), & \text{if } F_{s1}(\cdot) = F_{\text{seg}}(\cdot). \end{cases} \tag{8}$$

During the initial network training, we trained the SPNet with the initial pixel-level pseudo labels as the supervised information, and its loss function can be expressed as

$$L_{\text{step 1}}(x_i, \overline{y_l}) = L_{\text{Joint}}^{\text{intial}}(x_i, \overline{y_l}) \tag{9}$$

where $L_{\text{Joint}}^{\text{intial}}(x_i, \overline{y_l})$ represents that we use the joint loss function (JLF) L_{Joint} for network optimization with the initial pixel-level pseudo labels as the supervision. $\overline{y_l}$ represents the initial labels.

In the NLC and iterative update stage, the SPNet was optimized with the initial pseudo labels and the corrected

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pixel-level pseudo labels together as the supervised information, and its loss function can be calculated as

$$L_{\text{step 2}}\left(x_i, \overline{y_l}, \overline{\overline{y_l}}\right) = L_{\text{Joint}}^{\text{intial}}(x_i, \overline{y_l}) + \lambda_2 L_{\text{Joint}}^{\text{update}}\left(x_i, \overline{\overline{y_l}}\right) \quad (10)$$

where $L_{\text{Joint}}^{\text{update}}(x_i, \overline{\overline{y_l}})$ represents that we use the JLF L_{Joint} for network optimization with the corrected pixel-level pseudo labels as the supervision and $\overline{\overline{y_l}}$ represents the corrected labels.

2) ESB Semantic Segmentation: We used the network obtained from the noise label correction and iterative update stage (Step 2 in Section III-B) as the training starting point, and the fully corrected pixel-level pseudo labels were used as supervised information to fine-tune the SPNet. At this time, the SPNet was optimized using the JLF. Once the SPNet was trained well, the output results of the regular semantic segmentation head were the final ESB segmentation results.

IV. Experiments

A. Implementation Details

1) Experimental Environment: Our study was implemented in the PyTorch framework and trained on a single NVIDIA GeForce RTX 2080Ti GPU with 12 GB of video memory.

2) Image Classification Network: In our study, considering that the RegNetY-4.0GF network [57] can substantially reduce computational costs while achieving a satisfactory classification accuracy, we used it as the backbone of our image classification network. To mitigate a risk of overfitting, the dropout layer with the dropout rate of 0.2 was added before the fully connected layer. To explore the robustness of the SRIWS, we analyzed the sensitivity for the classification network backbone in Section V-D.

The image classification network was initialized using the pretrained weights trained on ImageNet. With the image-level labels as supervision, the network was trained using the SGD optimizer and cross-entropy loss. The process of training was performed for ten epochs with the learning rate of 0.001.

When the initial pixel-level pseudo labels were obtained from the CAM, t_1 and t_2 were set to 0.35 and 0.1, respectively.

3) SPNet Segmentation Network: It proposed in this article took the U-Net encoder–decoder architecture [58] to be the main body. The RegNetY-4.0GF was used to serve as the backbone of the encoder and the original U-Net continued to be the decoder. To explore the robustness of the SRIWS, we analyzed the backbone sensitivity for the SPNet in Section V-D.

During the initial network training stage, the SPNet was trained with a learning rate of 0.001 for two epochs using (9) as the optimization criterion. The hyperparameter λ_1 of the JLF $L_{\text{step 1}}(x_i, \overline{y_l})$ was set to 0.05. During the NLC and iterative update stage, the number of training epochs was set to 13 and the learning rate was fixed at 0.0001. The hyperparameters λ_1 and λ_2 of the JLF $L_{\text{step 2}}(x_i, \overline{y_l}, \overline{\overline{y_l}})$ were set to 0.05 and 1, respectively. During the ESB semantic segmentation stage, the process of training was performed for ten epochs with a learning rate of 0.001. The hyperparameter λ_1 was set to 0.2. The SPNet was trained with the Adam optimizer throughout all three stages.

4) SOTA Methods: The comparative experiments were conducted qualitatively and quantitatively with five state-of-the-art weakly supervised semantic segmentation approaches.

a) Coarse-to-fine weakly supervised segmentation method (CFWS) [31]: The coarse-to-fine weakly supervised segmentation method (CFWS) used superpixel segmentation results to constrain the initial semantic segmentation labels of the ground object. It corrected the noisy labels and performed boundary-aware segmentation to recognize the object extent.

b) Interpixel relation network [59]: The interpixel relation network (IRNet) utilized the vector relationship between pixel coordinates to detect the object extent and the semantic relationship between pixels to identify the object boundary, resulting in relatively complete object extraction results.

c) Deep seeded region growing [60]: The deep seeded region growing (DSRG) selected the most distinguishable sparse regions as the initial seed regions. It iteratively expanded the seed regions using a semantic segmentation network until relatively complete object extents were covered.

d) Two-step weakly supervised segmentation [22]: The two-step weakly supervised segmentation (TSWS) utilized the fixed threshold method and CRF operation to generate initial labels for ground objects. In addition, it introduced a CRF loss function into the segmentation network to improve spatial information.

e) Boundary exploration-based segmentation [61]: The boundary exploration-based segmentation (BES) employed CAM to generate boundary labels to train the boundary prediction network. Subsequently, the boundary prediction maps obtained from the boundary prediction network were served as constraints for the random walk, and the objects were then obtained by the segmentation network.

For a fair comparison, we performed the CFWS, IRNet, DSRG, TSWS, and BES using the same classification network (i.e., the RegNetY-4.0GF) and the semantic segmentation network (i.e., the U-Net) as implemented in the SRIWS.

B. Results of ESB Extraction

1) Accuracy Metrics: We selected six commonly used accuracy metrics [19], [21] to investigate the performance of the SRIWS for ESB recognition. The accuracy metrics are overall accuracy (OA), average intersection over union (mIoU), intersection over union (IoU) of ESB, F1 score of ESB, producer accuracy (PA) of ESB, and user accuracy (UA) of ESB.

2) Experiments on GE-ESB and GF2-ESB: The tests were performed on the image segmentation dataset of GE-ESB and GF2-ESB, and their corresponding results are shown in Table II. As shown in Table II, the SRIWS achieves the optimum accuracy for ESB semantic segmentation on GE-ESB and GF2-ESB, in comparison with the other five weakly supervised segmentation approaches. For the GE-ESB dataset, the SRIWS improves OA by 1.2%–4.1%. The IoU values and F1 score of ESB are increased by 4.1%–12.6% and 2.6%–8.4%, respectively. For the GF2-ESB dataset, the SRIWS shows an increase in OA by 1.1%–4.7%. The IoU values and F1 score of ESB are increased by 2.5%–12.2% and 1.9%–9.5%, respectively. In addition, the CFWS achieves the highest PA, while the TSWS has the highest UA on the GE-ESB dataset. On the GF2-ESB dataset, the PA of the IRNet is the highest, and the BES has the highest UA. Combining with the results

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TABLE II
ACCURACIES DERIVED FROM OUR SRIWS AND OTHER COMPETITORS ON THE IMAGE SEGMENTATION DATASET OF GE-ESB AND GF2-ESB

	Method	OA	mIoU	IoU	F1	UA	PA
GE-ESB	CFWS	0.900	0.785	0.700	0.823	0.724	**0.954**
	DSRG	0.911	0.801	0.717	0.835	0.757	0.932
	TSWS	0.912	0.778	0.663	0.798	**0.900**	0.716
	IRNet	0.922	0.823	0.746	0.854	0.786	0.936
	BES	0.929	0.830	0.748	0.856	0.848	0.865
	SRIWS	**0.941**	**0.857**	**0.789**	**0.882**	0.865	0.900
GF2-ESB	CFWS	0.884	0.719	0.575	0.730	0.649	0.836
	DSRG	0.887	0.728	0.592	0.744	0.647	0.875
	TSWS	0.903	0.723	0.556	0.714	0.799	0.647
	IRNet	0.910	0.773	0.653	0.790	0.703	**0.902**
	BES	0.920	0.758	0.606	0.755	**0.885**	0.658
	SRIWS	**0.931**	**0.799**	**0.678**	**0.809**	0.839	0.781

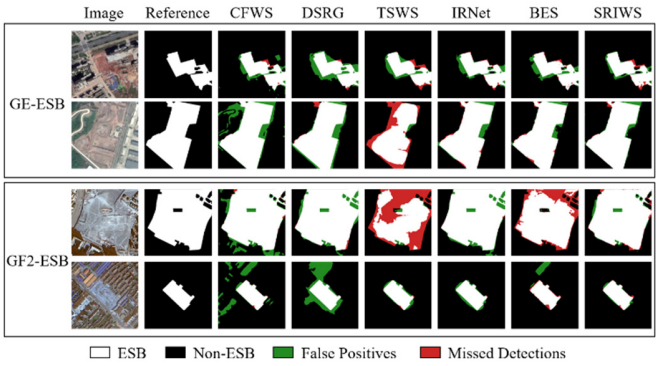

Fig. 8. Detailed information visualization of ESB extraction results derived from our SRIWS and other competitors on GE-ESB and GF2-ESB.

shown in Fig. 8, it appears that these methods exhibit varying degrees of false positives and omissions. As shown in Fig. 8, the shape of ESB is irregular and the size varies greatly, which poses challenges for accurately detecting the ESB extents with just image-level labels. However, in comparison to the other five weakly supervised approaches that introduce false positives and missed detections to varying degrees, the SRIWS demonstrates the capability to extract more complete ESB on both datasets while effectively preserving the boundaries. This observation highlights the effectiveness and universality of the SRIWS.

Our proposed SRIWS has similar functions to CFWS, but the accuracy varies greatly for the following reasons.

First, during the training of the image classification network, SRIWS utilized SRMix to generate more accurate and complete CAMs (as shown in Section V-A). In addition, with the same backbone setting, the reasons for the lack of accuracy in CFWS are explained through the three steps included in both CFWS and SRIWS.

a) Initial pixel-level pseudo labels generation: Both CFWS and SRIWS use the graph-based image segmentation algorithm [53] and unsupervised image segmentation technique [62] to generate superpixels. These superpixels can effectively constrain the object boundary and provide prior knowledge of object shape. However, some noise is present in these superpixels, as shown in Fig. 9. When generating initial pixel-level pseudo labels using CAM, CFWS employs an object-based approach, which may introduce noise from the superpixels into the pseudo labels. On the contrary, the initial pixel-level pseudo labels produced by SRIWS perform well in terms of quality metrics (as shown in Section V-A).

b) Pseudo label correction: When using the initial pseudo labels for noise label correction, CFWS faces challenges as its pseudo labels contain more noise. This can lead the segmentation network to mistakenly include noise objects as part of the target object during correction, affecting the quality of the corrected labels, as shown in Fig. 9. In contrast, SRIWS introduces a superpixel pooling layer during the label noise correction and performs an HSF based on the output of the network. Using the spatial information provided by the superpixels and the semantic information extracted from the network itself, SRIWS classifies the uncertain pixels and corrects the noise in the pseudo labels, resulting in the corrected labels with good quality and fine boundaries.

c) ESB semantic segmentation: The segmentation network of CFWS is affected by the noise in the fully corrected labels, leading to more false positives in the segmentation results. The SRIWS optimizes the SPNet using the JLF, which makes full use of the spatial information of the superpixels, thus extracting a more complete and accurate range of ESBs.

Among the complex and diverse backgrounds, low-cover grassland exhibits sparse grass coverage and may have large areas of bare soil. Impervious surface areas (ISA), which may share similar spectral characteristics with ESB, generally originate from ESB and are spatially close to it. Furthermore, crop rotation and fallow land may display substantial patches of bare soil during the fallow period. The above ground objects introduce challenges to the extraction of ESB. In this regard, we used manual interpretation to obtain pixel-level labels corresponding to low-coverage grassland, ISA, and cropland. Subsequently, we visualized and analyzed these confusing regions.

i) ESB mixed with low-coverage grassland: Fig. 10 shows that the SRIWS outperforms other methods in capturing the distinction between low-coverage grassland and ESB, resulting in better segmentation performance. Conversely, the CFWS and the DSRG are easy to confuse low-coverage grassland and ESB, leading to the introduction of numerous false positives during the ESB extraction. The TSWS has a high level of omissions on the GE-ESB dataset and many false positives on the GF2-ESB dataset. The IRNet and the BES have some ability to distinguish between low-coverage grassland and ESB but perform poorly on the GF2-ESB dataset.

ii) ESB mixed with ISA: Fig. 11 shows that the SRIWS exhibits the best performance in distinguishing ESB from ISA. However, due to the spatial adjacency and characteristics between ISA and ESB, the SRIWS introduces a few false positives caused by the misclassification of ISA. The other five methods also misclassify ISA as ESB to different degrees

iii) ESB mixed with cropland: From Fig. 12, it is evident that the SRIWS has the strongest ability to classify between cropland and ESB. In contrast, the CFWS exhibits the weakest ability to distinguish between cropland and ESB, resulting in misclassifications. The DSRG, TSWS, IRNet, and BES

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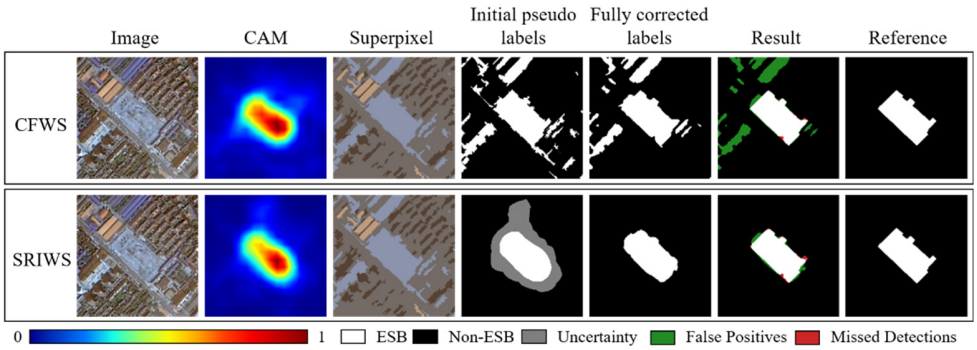

Fig. 9. Intermediate results of CFWS and SRIWS. The legend for CAM, labels, results, and references is located at the bottom of the figure. In superpixel maps, pixels of the same color belong to the same superpixel, and the pixel color is the average of the spectral values of all pixels in the image within the superpixel.

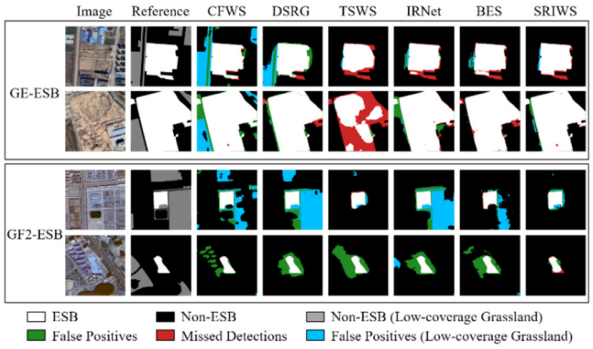

Fig. 10. Detailed information visualization of the extraction results for the ESB mixed with low-coverage grassland derived from our SRIWS and other competitors on GE-ESB and GF2-ESB. We use gray regions to present the label of low-coverage grassland.

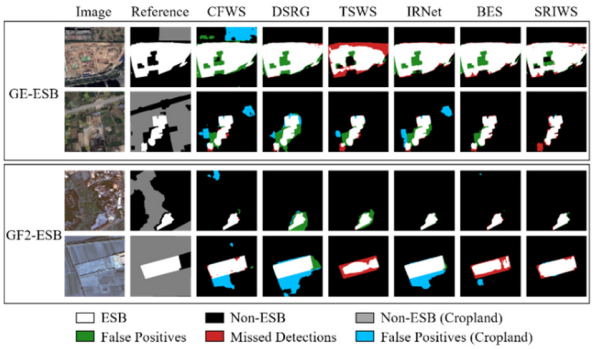

Fig. 12. Detailed information visualization of the extraction results for the ESB mixed with cropland derived from our SRIWS and other competitors on GE-ESB and GF2-ESB. We use gray regions to present the label of cropland.

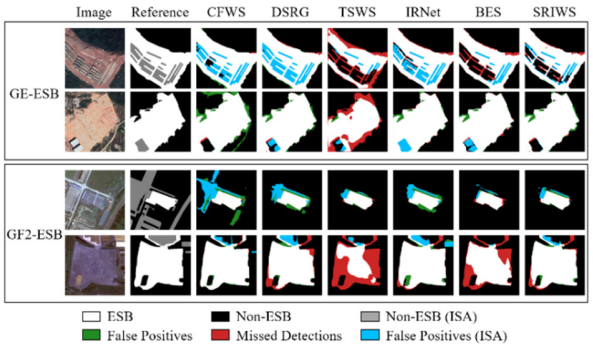

Fig. 11. Detailed information visualization of the extraction results for the ESB mixed with ISA derived from our SRIWS and other competitors on GE-ESB and GF2-ESB. We use gray regions to present the ISA label.

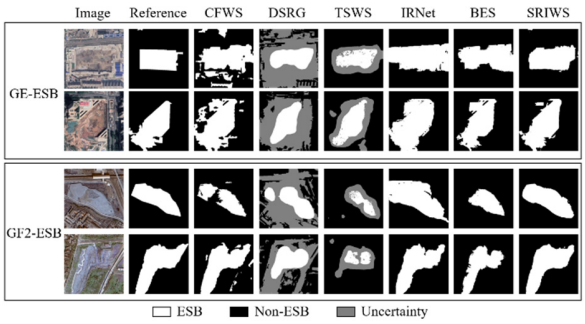

Fig. 13. Visualization of pseudo labels derived from our SRIWS and other methods on the GE-ESB and GF2-ESB datasets.

have some ability to classify between the two ground objects. However, they are easy to confuse cropland and ESB when the spectral characteristics of the two ground objects are similar. In addition, the omission of TSWS is somewhat more serious.

To further validate the effectiveness of SRIWS, we visualized the pseudo labels generated by our SRIWS and other weakly supervised methods, as shown in Fig. 13. It is evident that the pseudo labels produced by the SRIWS exhibit greater accuracy and are closer to the references, presenting fewer false positives and omissions in contrast to the other weakly supervised methods.

3) Test in the Scenarios With Sparse and Unbalanced ESB Samples: We performed tests on the large-area (city-level) test dataset to evaluate various weakly supervised segmentation approaches. The quantitative results are presented in Table III, and Fig. 14 shows the detailed information visualization. The tests show that the SRIWS extracts more complete ESB with more refined boundaries while introducing fewer false positives and achieving superior performance on the large-area (city-level) test dataset, compared to other methods.

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TABLE III

ACCURACIES DERIVED FROM OUR SRIWS AND OTHER COMPETITORS ON THE LARGE-AREA (CITY-LEVEL) TEST DATASET

	Method	OA	mIoU	IoU	F1	UA	PA
Xiaogan	CFWS	0.982	0.774	0.566	0.723	0.661	0.798
	DSRG	0.969	0.706	0.444	0.615	0.485	**0.843**
	TSWS	0.985	0.782	0.579	0.733	0.748	0.719
	IRNet	0.981	0.769	0.558	0.716	0.640	0.814
	BES	0.982	0.778	0.574	0.730	0.667	0.806
	SRIWS	**0.987**	**0.811**	**0.635**	**0.777**	**0.771**	0.783
Foshan	CFWS	0.991	0.778	0.565	0.722	0.643	0.824
	DSRG	0.986	0.728	0.471	0.640	0.504	**0.876**
	TSWS	0.992	0.771	0.550	0.710	0.753	0.671
	IRNet	0.992	0.796	0.599	0.749	0.686	0.825
	BES	0.992	0.790	0.589	0.741	0.700	0.787
	SRIWS	**0.993**	**0.802**	**0.610**	**0.758**	**0.807**	0.714

V. DISCUSSION

A. Effectiveness of SRMix

To verify the effectiveness of SRMix, we evaluated the quality of the initial pixel-level pseudo labels. We selected two metrics, i.e., accuracy and completeness, to calculate the quality evaluation metric Qua. Notably, when calculating the accuracy metric, we ignored the uncertain pixels in the initial pixel-level pseudo labels. Qua is the harmonic mean of the accuracy and the completeness metric [63]

$$\text{Comple} = \frac{\text{Num}_{\text{ESB}} + \text{Num}_{\text{Non-ESB}}}{\text{Num}_{\text{ESB}} + \text{Num}_{\text{Non-ESB}} + \text{Num}_{\text{Uncertain}}} \tag{11}$$

$$\text{Qua} = \frac{(1 + \text{beta}) * \text{mIoU} * \text{Comple}}{\text{beta} * \text{mIoU} + \text{Comple}} \tag{12}$$

in which Num_{ESB} represents the number of ESB pixels in the initial pixel-level label, $\text{Num}_{\text{Non-ESB}}$ represents the number of Non-ESB pixels, and $\text{Num}_{\text{Uncertain}}$ represents the number of uncertain pixels. The beta represents the weight, and a higher value for beta indicates that the accuracy metrics are given more importance than the quality metrics. In this section, the value of beta is set to 1.

1) Comparison of the SRMix With Other Data Augmentation Methods: Using the data augmentation method, the image classification network was trained to generate initial pixel-level pseudo labels. Then, we evaluated the quality of these labels on the image segmentation datasets. For the sake of testing the performance of our SRMix, we performed a comparison with four commonly used data augmentation methods, namely, the MixUp [64], the CutMix [65], the FMix [66], and the semantically proportional mixing (SnapMix) [49].

From the perspective of image values, MixUp linearly interpolates pairs of images and uses the same coefficients to fuse their corresponding labels, resulting in the generation of new pseudo samples. Based on the image space, CutMix cuts and pastes a random rectangular patch among images while proportionally mixing their corresponding labels in terms of the area composition. Improved from the CutMix, the FMix uses a random binary mask acquired by thresholding the low-frequency images derived from Fourier space to generate pseudo images. Compared with the CutMix, the FMix converts the cropped patch from the original rectangle to an irregular shape. The SnapMix cuts a random patch and pastes it onto another image, while their pseudo labels are obtained by estimating its intrinsic semantic composition.

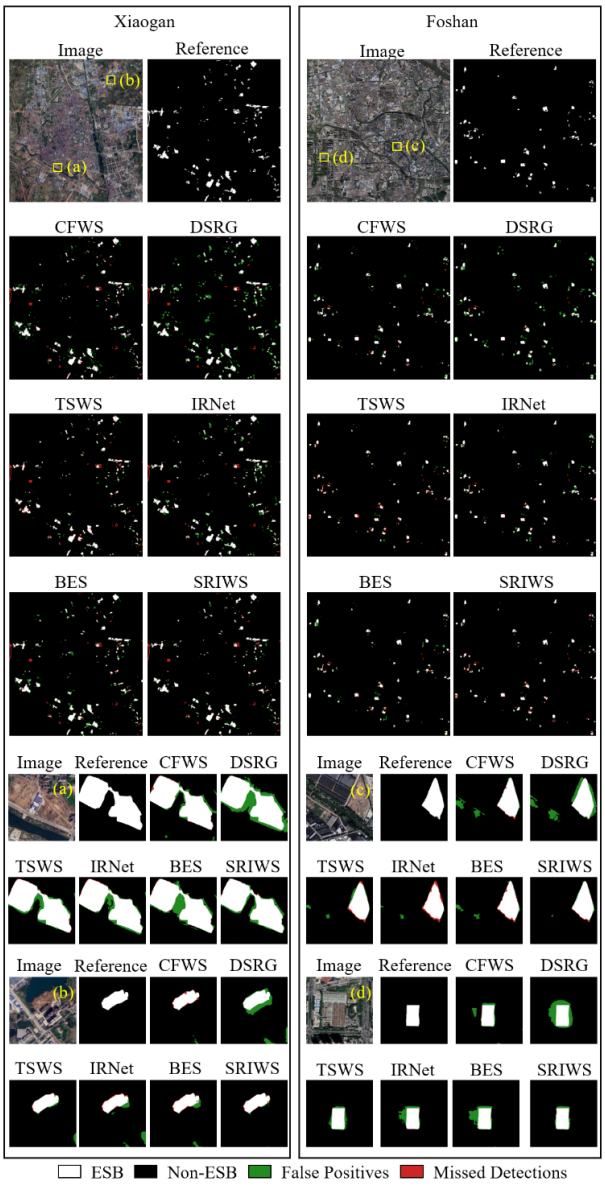

Fig. 14. Detailed information visualization of ESB extraction results derived from our SRIWS and other competitors on the large-area (city-level) test dataset. (a)–(d) Display zoomed-in views of selected areas in Xiaogan and Foshan, with each view having an image size of 512 × 512 pixels.

The quality of the initial pixel-level pseudo labels generated by the various data augmentation methods is shown in Fig. 15. Fig. 15 shows that the use of SRMix in the image classification network results in more accurate and complete initial pixel-level pseudo labels compared to other data augmentation methods. MixUp is relatively effective on the GF2-ESB dataset but does not play a significant effect on the GE-ESB dataset. FMix and SnapMix are effective on the GF2-ESB dataset but do not work well on the GE-ESB dataset. The CutMix performs poorly on both datasets.

The poor performance of the CutMix, FMix, and SnapMix may be attributed to the factor that they generate new pseudo-samples based on the image space while using soft labels to calculate the pseudo labels so that a certain amount

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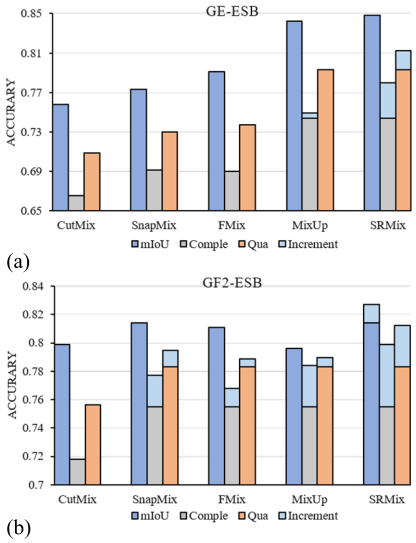

Fig. 15. Quality of the initial pixel-level pseudo labels generated by the various data augmentation methods. Baseline is the results without data augmentation, and the increment refers to the increase against the baseline. Results on (a) GE-ESB segmentation dataset and (b) GF2-ESB segmentation dataset.

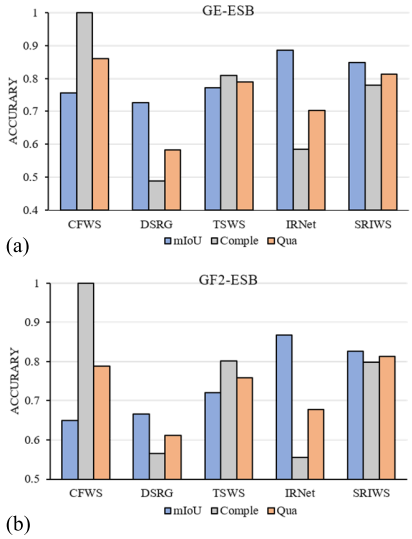

Fig. 16. Quality of the initial pixel-level pseudo labels generated by our SRIWS and other competitors. Results on (a) GE-ESB segmentation dataset and (b) GF2-ESB segmentation dataset.

of noise is introduced during the sample synthesis. When the hindering effect of noise samples on model training outweighs the benefits of sample diversity, it can negatively affect the performance of the CAMs to some extent.

2) Comparison With the State-of-the-Art Weakly Supervised Methods: In this article, we have conducted comparative experiments with five weakly supervised methods, i.e., CFWS, DSRG, TSWS, IRNet, and BES, to evaluate the effectiveness of SRIWS. Since the initial pixel-level pseudo labels generated by the BES contain boundary pixels, we evaluate the quality of the initial pixel-level pseudo labels generated by the four weakly supervised methods (CFWS, DSRG, TSWS, and IRNet) in this section, and the results are shown in Fig. 16.

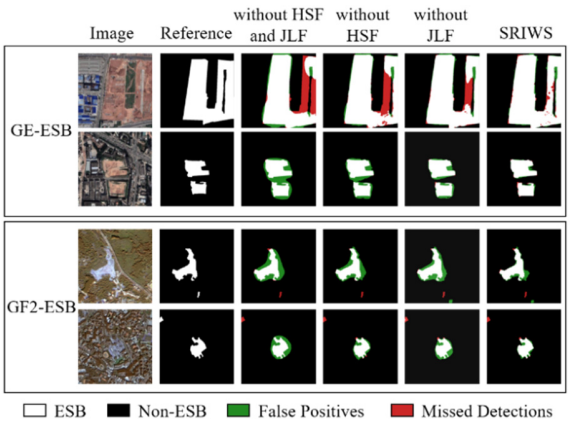

Fig. 17. Ablation study results for the SPNet on the GE-ESB and GF2-ESB datasets. JLF: joint loss function. HSF: high-confidence sample filtering operation.

From Fig. 16, it can be seen that the initial pixel-level pseudo labels generated by our SRIWS exhibit the highest quality on the GF2-ESB dataset. On the GE-ESB dataset, the pseudo labels generated by SRIWS surpass the other methods in terms of quality metrics, except for CFWS. It is important to note that CFWS employs the Otsu threshold method to generate the initial pixel-level pseudo labels, resulting in complete but less accurate pseudo labels. In contrast, our SRIWS has a better balance between accuracy and completeness.

B. Ablation Study for the SPNet

1) Joint Loss Function: To investigate the effectiveness of the JLF [as shown in (5)], we compare it with the case without the operation, i.e., two branches of the SPNet are optimized with cross-entropy loss separately, as shown in (6). The accuracy results on the GE-ESB and GF2-ESB datasets are presented in Table IV (see the fifth and sixth rows), and the visualization information is shown in Fig. 17 (see the fifth and sixth columns). Combining the results from Table IV with Fig. 17, it becomes evident that the JLF plays a beneficial role in the SPNet training. It helps to capture details provided by the superpixels, mitigate false positives and omissions, and better preserve the ESB boundaries.

2) High-Confidence Sample Filtering Operation: To investigate the effectiveness of the HSF, we compared the results with and without this operation on GE-ESB and GF2-ESB. The accuracies are presented in Table IV (see the fourth and sixth rows), and the visualization information is shown in Fig. 17 (see the fourth and sixth columns). Omitting the use of HSF during the NLC stage implies that using the predictions of the regular semantic segmentation head as the corrected pixel-level pseudo labels in Step 2 of Section III-B. From Fig. 18, we can clearly observe the positive effect of the HSF on the NLC stage. Combining the accuracy results in Table IV with the visualization results in Figs. 17 and 18, it is obvious that our HSF helps the SPNet to learn from purer samples, resulting in more accurate and refined ESB labels during the NLC stage and thus improving the overall ESB semantic segmentation accuracy.

3) Noisy Label Correction: For the sake of investigating the performance of the NLC technique, a comparative analysis

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TABLE IV

ABLATION STUDY FOR THE SPNET. JLF: JOINT LOSS FUNCTION. HSF: HIGH-CONFIDENCE SAMPLE FILTERING OPERATION. NLC: NOISY LABEL CORRECTION. f_{seg}: REGULAR SEMANTIC SEGMENTATION HEAD. f_{sup}: SEMANTIC SEGMENTATION HEAD WITH A SUPERPIXEL POOLING LAYER

Dataset	JLF	HSF	NLC	Semantic segmentation heads		OA	mIoU	IoU	F1	UA	PA
				f_{seg}	f_{sup}						
GE-ESB			√	√		0.902	0.779	0.682	0.811	0.766	0.861
			√	√	√	0.908	0.789	0.694	0.820	0.780	0.863
	√			√	√	0.910	0.791	0.695	0.819	0.799	0.842
	√		√	√	√	0.925	0.822	0.739	0.850	0.830	0.871
		√	√	√	√	0.938	0.849	0.778	0.875	0.855	0.897
	√	√	√	√	√	**0.941**	**0.857**	**0.789**	**0.882**	**0.865**	**0.900**
GF2-ESB			√	√		0.895	0.742	0.610	0.758	0.667	**0.878**
			√	√	√	0.906	0.755	0.622	0.767	0.715	0.828
	√			√	√	0.915	0.768	0.637	0.778	0.757	0.801
	√		√	√	√	0.920	0.781	0.655	0.792	0.774	0.810
		√	√	√	√	0.925	0.790	0.668	0.801	0.798	0.807
	√	√	√	√	√	**0.931**	**0.799**	**0.678**	**0.809**	**0.839**	0.781

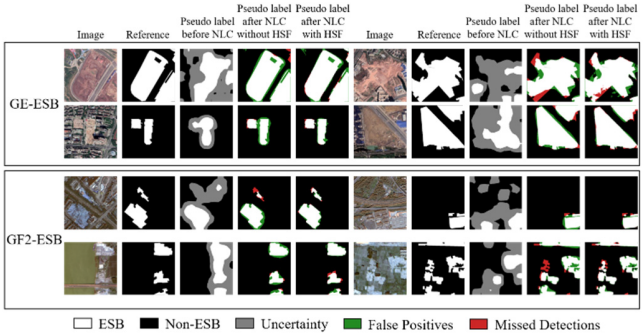

Fig. 18. Pixel-level pseudo label before and after the NLC. HSF: high-confidence sample filtering operation. NLC: noisy label correction.

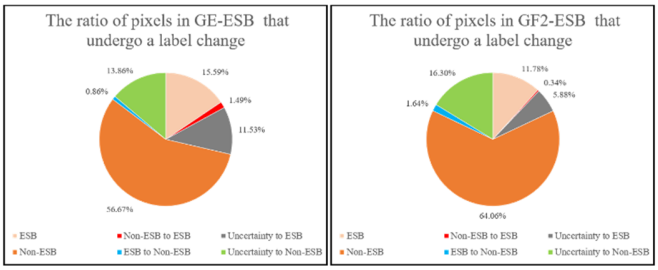

Fig. 19. Ratio of pixels that undergo a label change from the initial pseudo labels to the fully corrected pseudo labels before and after the NLC technique.

between the experiments with and without the operation was conducted on GE-ESB and GF2-ESB. Without the NLC, the SPNet is unable to perform the HSF. The NLC technique has the ability to mitigate the false positives in the initial pixel-level pseudo labels and obtain more accurate and complete ESB pseudo labels. This improvement can be observed in Table IV (third and fourth rows) and Fig. 18, where the extent of ESB pseudo labels is more refined after the NLC. In addition, when comparing the experimental results to those obtained without the HSF, it can be observed that the NLC technique contributes to improving the final segmentation accuracies of the SPNet to some extent.

For the sake of further investigation on the influence of the NLC technique, we calculated the percentage of transformed pixels that undergo a change in label for all the samples before and after the NLC technique. As shown in Fig. 19, for the GE-ESB dataset, the NLC transformed 1.49% of Non-ESB pixels in the initial pseudo labels into ESB pixels, 11.53% of uncertain pixels into ESB pixels, 0.86% of ESB pixels into Non-ESB pixels, and 13.86% of uncertain pixels into Non-ESB pixels. For the GF2-ESB dataset, the NLC transformed 0.34% of Non-ESB pixels into ESB pixels, 5.88% of uncertain pixels into ESB pixels, 1.64% of ESB pixels into Non-ESB pixels, and 16.30% of uncertain pixels into Non-ESB pixels. These findings demonstrate the significant impact of the NLC in refining and denoising the initial pseudo labels.

4) Superpixel Pooling Layer: For the sake of evaluating the effect of the superpixel pooling layer in the SPNet, we conducted experiments using single-branch networks on GE-ESB and GF2-ESB, and the accuracies are presented in Table IV (see the first and second rows). When the deep learning network contains only single branch, the JLF as well as the HSF cannot be used. The results in Table IV show that the utilization of the superpixel pooling layer facilitates our SPNet to learn the spatial information supplied by the superpixels and raise the ESB detection accuracies. However, to fully utilize the spatial details brought by the superpixels and further improve accuracy, the addition of the JSF and the HSF is needed.

C. Comparison of the Model Efficiency

The model efficiency of different weakly supervised methods is compared in Table V. We followed the same settings for the CFWS, IRNet, DSRG, TSWS, and BES according to the image classification network (RegNetY-4.0GF) and the image segmentation network (U-Net) used by our SRIWS, resulting in a similar number of model parameters (24.73 MB). The test was implemented in the PyTorch framework and trained on a single NVIDIA GeForce RTX 2080Ti GPU with 12 GB of video memory. As shown in Table V, the model efficiencies of the six weakly supervised methods do not differ much due to the consistent backbone used by these methods. Notably, IRNet and BES, due to its augmentation of the input image with horizontal and vertical flipping, exhibit higher giga multiply–accumulate operations per second (GMACs).

In addition, we calculated the training time for each method. As shown in Fig. 20, we note that IRNet and BES require

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TABLE V

MODEL EFFICIENCY OF OUR SRIWS AND OTHER COMPETITORS. GMACS: GIGA MULTIPLY–ACCUMULATE OPERATIONS PER SECOND. FPS: FRAMES PER SECOND. S.: SECOND

		CFWS [31]	DSRG	TSWS [22]	IRNet [59]	BES [61]	SRIWS
Total	GMACs	35.12	35.18	35.16	48.86	50.05	35.16
	FPS (s.)	0.103	0.058	0.064	0.076	0.080	0.109

TABLE VI

ACCURACIES OF THE SRIWS ON GE-ESB AND GF2-ESB UNDER DIFFERENT BACKBONES

Network	Dataset	Backbone	Parameters	OA	mIoU	IoU	F1	UA	PA
The image classification network	GE-ESB	RegNetY-4.0GF	24,727,836	**0.941**	**0.857**	**0.789**	**0.882**	**0.865**	**0.900**
		ResNet34	24,436,548	0.939	0.854	0.786	0.880	0.841	0.924
		VGG16_bn	23,756,868	0.940	0.855	0.788	0.881	0.842	0.924
	GF2-ESB	RegNetY-4.0GF	24,727,836	**0.931**	**0.799**	**0.678**	**0.809**	**0.839**	**0.781**
		ResNet34	24,436,548	0.928	0.791	0.667	0.800	0.835	0.770
		VGG16_bn	23,756,868	0.929	0.795	0.674	0.806	0.825	0.787
The SPNet	GE-ESB	RegNetY-4.0GF	24,727,836	**0.941**	**0.857**	**0.789**	**0.882**	**0.865**	**0.900**
		ResNet34	24,436,548	0.936	0.845	0.773	0.872	0.847	0.899
		VGG16_bn	23,756,868	0.935	0.843	0.769	0.869	0.848	0.892
	GF2-ESB	RegNetY-4.0GF	24,727,836	**0.931**	**0.799**	**0.678**	**0.809**	**0.839**	**0.781**
		ResNet34	24,436,548	0.923	0.772	0.633	0.775	0.856	0.708
		VGG16_bn	23,756,868	0.925	0.782	0.651	0.789	0.837	0.746

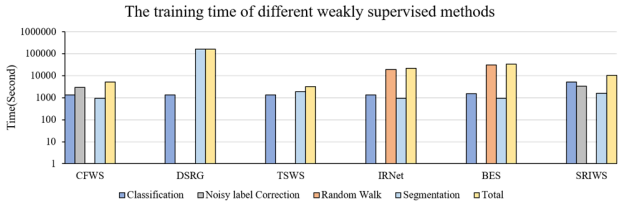

Fig. 20. Training time of our SRIWS and other competitors.

additional networks for random walk. Overall, our SRIWS achieves the best segmentation performance with a moderate training time.

D. Sensitivity Analysis for the Backbone of the SRIWS

To test the robustness of SRIWS, we replaced the backbone of the image classification network and the SPNet in turn while keeping other settings unchanged. The accuracies on GE-ESB and GF2-ESB are presented in Table VI.

VGGNet [67] is a classical deep learning model that achieves high performance by adding the convolutional layers to increase the depth while reducing the size of convolutional filters. VGGNet is known for its simplicity and effectiveness in feature extraction from images. It is a widely used backbone in deep learning that is applied in various applications [24], [68]. ResNet [69] is a classical architecture in deep learning models, which tackles the degradation of deep networks through the residual learning framework. Most of the existing studies used ResNet as the backbone due to its simple structure and diverse variants [25], [33]. To ensure a fair comparison of performance, we chose the VGG16_bn network and the ResNet34 network as a backbone. These networks have a similar number of parameters to the RegNetY-4.0GF.

Table VI shows that the SRIWS can be still trained well and achieve great performance using different backbones.

According to [57], under the same conditions, RegNetY generally outperforms RegNetX due to the squeeze and excitation (SE) structure. Furthermore, RegNetX models are obtained by progressively optimizing the CNN paradigms and hence usually outperform ResNet under the same training settings. According to [69], the residual structure in ResNet can prevent the degradation of deep networks and achieve accuracy gains from network depth, which generally results in better performance than VGGNet. Therefore, there may be some accuracy degradation when replacing the RegNetY-4.0GF network with the ResNet34 and VGG16 networks.

Considering that several differences exist in the accuracies of the SRIWS under various backbones, we selected the RegNetY-4.0GF network as the backbone in the previous presentation of results.

VI. CONCLUSION

Our study concentrates on weakly supervised learning with image-level labels for ESB extraction from the VHR images, which is crucial for planning urban development and improving the urban environment. We propose a SRIWS. The SRIWS first trains the image classification network using image-level samples to produce initial pixel-level pseudo labels. During the image classification network training, the SRIWS use an oversampling strategy, termed SRMix, to oversample the ESB samples and effectively alleviate the influence of class imbalance. Subsequently, the SRIWS uses a two-branch SPNet for iterative correction of the initial pseudo labels. Finally, the SRIWS continues to fine-tune the SPNet with the fully corrected pixel-level pseudo labels as supervised information to obtain the overall ESB extraction results.

We tested the SRIWS on three datasets, that is, GE-ESB, GF2-ESB, and large-area (city-level) test datasets. The experimental results clearly indicate that our SRIWS is superior to the five existing state-of-the-art weakly supervised approaches. It achieves higher accuracy and extracts more complete ESB extents with effectively retaining boundaries on all three datasets. In addition, we find that the SRMix can generate more accurate and complete CAMs compared to the four commonly used data augmentation methods and the weakly supervised methods. Our ablation experiments on

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the SPNet segmentation network confirm that the JLF, high-confidence sample filtering operation, noise label correction, and superpixel pooling layer are all beneficial to improve the performance of the SPNet. These components enable the SPNet to effectively preserve both global information and spatial details provided by superpixels, leading to more precise ESB segmentation results with refined boundaries. Finally, it is observed that the performance of the SRIWS is not heavily reliant on the specific backbone of the image classification network and the SPNet, which demonstrates the robustness and university of the SRIWS.

There are still some limitations in our study. First, the current noise label correction operation relies on manual parameter setting and lacks automatic termination during network training. In this regard, we consider setting appropriate automatic termination conditions to reduce the manual intervention. Second, while the SRIWS has demonstrated strong performance in ESB extraction, there are still instances where false positives and omissions occur, especially in scenes with sparse and unbalanced ESB samples. In future research, we consider adding additional features, such as semantic affinity, to boost the ability of the network to recognize ESB.

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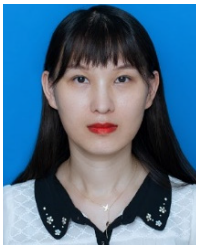
Xin Huang (Senior Member, IEEE) received the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2009.

He is with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University, where he is currently a Full Professor and teaches remote sensing and image interpretation. He is the Head of the Institute of Remote Sensing Information Processing (IRSIP), School of Remote Sensing and Information Engineering, Wuhan University. He has been supported by the National Program for Support of Top-notch Young Professionals in 2017, the China National Science Fund for Excellent Young Scholars in 2015, and the New Century Excellent Talents in University from the Ministry of Education of China in 2011. He has published more than 200 peer-reviewed articles (Science Citation Index (SCI) articles) in international journals. His research interests include remote sensing image processing methods and applications.

Prof. Huang was a recipient of the Boeing Award for the Best Paper in Image Analysis and Interpretation from the American Society for Photogrammetry and Remote Sensing (ASPRS) in 2010, the John I. Davidson President's Award from ASPRS in 2018, and the National Excellent Doctoral Dissertation Award of China in 2012. He was the winner of the IEEE GRSS Data Fusion Contest in 2014 and 2021. He was an Associate Editor of the *Photogrammetric Engineering and Remote Sensing* from 2016 to 2019 and an Associate Editor of the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS from 2014 to 2020 and the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING from 2018 to 2022. He has been serving as an Associate Editor for IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING since 2022. He has been serving as an Editorial Board Member for the *Remote Sensing of Environment* since 2019.

Wenrui Wang received the B.S. degree in remote sensing from Wuhan University, Wuhan, China, in 2021, where she is currently pursuing the Ph.D. degree in photogrammetry and remote sensing with the School of Remote Sensing and Information Engineering.

Her research interests include land-cover classification, semantic segmentation, high-resolution image processing, and deep learning.



Jiayi Li (Senior Member, IEEE) received the B.S. degree from Central South University, Changsha, China, in 2011, and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2016.

She is currently an Associate Professor with the School of Remote Sensing and Information Engineering, Wuhan University. She has authored more than 60 peer-reviewed articles (Science Citation Index (SCI) articles) in international journals. Her research interests include hyperspectral imagery, sparse representation, computation vision and pattern recognition, and remote sensing images.

Dr. Li is a reviewer for more than 30 international journals, including IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE TRANSACTIONS ON CYBERNETICS, RSE, and ISPRS-J. She is a Young Editorial Board Member of *Geo-Spatial Information Science* (GSIS) and the Guest Editor of the *Remote Sensing* (an open access journal from MDPI) and *Sustainability* (an open access journal from MDPI).

Leiguang Wang received the Ph.D. degree from the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University, Wuhan, Hubei, China, in 2009.

From 2014 to 2015, he was a Post-Doctoral Researcher with the University of New Brunswick, Fredericton, NB, Canada. He is currently a Professor with Southwest Forestry University, Kunming, China, where he has been the Vice Dean of the Institute of Big Data and Artificial Intelligence since 2018. He is the author of more than 50 articles. His research interests include remote sensing image fusion, semantic segmentation, and application in forestry.

Xing Xie is with the School of Environmental and Mapping Engineering, Suzhou University, Suzhou, China. His research interests include Gabor filters, image classification, image fusion, image segmentation, image texture, and parameter estimation.

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